

Pre-Art: A Categorical and Homotopy-Theoretic Framework for Art, Toward Universal Expressionism

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Abstract

We propose a formal category theoretic and homotopy type theoretic framework for understanding art. Rather than attempting to define art, an undecidable question, we introduce the notion of *pre-art*: morphisms between substrate states from which all artistic composition is built. Pre-arts form a nonabelian free group $F(\mathcal{P})$, with elements as compositions. A *context* is any subset of $F(\mathcal{P})$ —an exhibition, a series, a wall. Design principles are constructed by collecting all compositions exhibiting a given structural property. Ideas are defined as cones over all their instantiations; design principles are a special case. Finally, we introduce *Universal Expressionism* as the artistic movement consisting of works in universally principled contexts—those in which every composition exhibits at least one design principle. Accessibility is built directly into the theory: anyone who can adjoin a universal pre-art to a universal context in a universally principled manner can participate, which can easily be achieved through simple shapes and iconographic forms built from elementary primitives. Participation is thus compositional rather than institutional.

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1 The Definitional Trap

Every formal theory of art must confront the following obstacle.

The Definitional Problem

Any proposed necessary and sufficient condition for “art” is falsified by some recognized artwork. Duchamp’s *Fountain* (1917) falsifies beauty conditions. Cage’s *4’33”* falsifies sound-production conditions. A Rauschenberg erased De Kooning drawing falsifies additive-process conditions. The history of art is precisely the history of such falsifications.

Existing mathematical treatments of art share a common flaw: they assume a pre-theoretic notion of “artwork” and build formal structure on top of it. The problem is that artworks are not a well-defined class. What counts as an artwork is contested, historically contingent, institutionally mediated, and perpetually renegotiated. Any definition of art is a hostage to the next provocation.

Our resolution is to *refuse the question*. We do not define art. We define something more primitive—pre-art—and observe that art, wherever it is found, is always composed of pre-art. The boundary between pre-art and art is not drawn by the theory. It is left to context, institution, and the judgment of participants. This is not a weakness; it is the correct formal response to an undecidable question.

2 Substrates

Definition 2.1 (Substrate). A *substrate* is any material, medium, or support upon which transformations may be performed.

Example 2.2. Canvas, linen, paper, photographic sensor, digital representation, stone, found object, human body, silence, wall, floor, space. Conceptual substrates are also possible—an idea, a design principle.

Definition 2.3 (Conceptual Substrate). A *conceptual substrate* is any idea, concept, or semantic domain upon which transformations may be performed. It is a substrate in the sense of Definition 2.1, with the material support replaced by a domain of meaning.

Example 2.4. Grief, duality, entropy, the void, the number seven, the color red as symbol rather than pigment, the institutional critique of the white cube—each is a conceptual substrate. The initial state $\emptyset_S$ of such a substrate is the concept in its latent, uninstantiated form: present in the culture but not yet embodied in a particular work.

Definition 2.5 (Initial State). Given a substrate S , the *initial state* $\emptyset_S$ is the starting point of the substrate as chosen by the artist.

Remark 2.6. Typically, the initial state will be the substrate’s “least-transformed”, most-available configuration, from the artist’s perspective: the empty canvas, the blank page, the unrecorded silence, the bare wall.

Remark 2.7. The initial state is always a readymade object. Note that the readymade object is a part of, but distinct from the readymade itself, which consists of the readymade object and one or more ideas adjoined to it. The idea and the readymade object are two distinct substrates. This observation places the framework in explicit dialogue with the Duchampian tradition: all art is, at some level of abstraction, found art.

In this way, a readymade is the identity morphism on its initial state. If we let the readymade object be X and the idea I , the initial and final state is $(\emptyset_X, \emptyset_I)$. Note that $\emptyset_X$ itself may be a composition $x_0 \circ \dots \circ x_i$ that is chosen to be initial, and likewise with $\emptyset_I$. The choice of initial is according to someone’s perspective and is thus arbitrary. The implication is that any artwork itself could reasonably be considered as a readymade.

Definition 2.8 (Substrate State Space). Let S be a substrate. The *state space* $\mathcal{S}(S)$ is the collection of all possible states of S under transformation.

Remark 2.9. Since any collection of substrates is a set, (co)products of substrates—sets of ordered pairs of substrates—are also naturally substrates. Typically, an artwork begins with some physical substrates such as canvas and paint, and with an idea (conceptual), so the initial substrate can be represented as an ordered pair (canvas, paint, idea) as an element of $\mathcal{S}(\text{Canvas}) \oplus \mathcal{S}(\text{Paint}) \oplus \mathcal{S}(\text{Idea})$.

3 Pre-Art as Morphisms

The central move of this framework is to model pre-art not as objects but as *morphisms*—transformations between states of a substrate.

Definition 3.1 (Pre-Art). A *pre-art entity*, henceforth pre-art, is a morphism

$$f : A \longrightarrow B$$

where $A, B \in \mathcal{S}(S)$ for some substrate S .

That is, a pre-art is any transformation that carries one state of a substrate to another.

Example 3.2. A mark, a gesture, a brushstroke, a cut, a choice of lighting, a digital filter, a found juxtaposition, a placement of an object in a room, the application of varnish, the hanging of a work on a wall, are all pre-arts.

Example 3.3. A pre-art may act on a conceptual substrate.

Remark 3.4. Notably, we take the category theoretic flavor of thinking about “objects” as functions (or actions) themselves, motivating the following:

Example 3.5. A substrate is also pre-art, as its own identity morphism $\mathbb{1}_S : S \rightarrow S; \forall s \in S, \mathbb{1}_S(s) = s$. Conversely, a pre-art is also a substrate upon which transformations may be performed. Hence, pre-arts and substrates are definitionally equivalent. Implied is a notion of choice of what is a substrate, and by extension, a pre-art.

Example 3.6. The readymade is also pre-art, as its own identity morphism $\mathbb{1}_{(\emptyset_X, \emptyset_I)} : (\emptyset_X, \emptyset_I) \rightarrow (\emptyset_X, \emptyset_I)$. Note that the readymade is again defined by the choice of an idea, object, and initial state, as substrates are generally already readymades as prefabricated compositions of arrows. However, a pre-art need not be a readymade, as a pre-art doesn’t necessarily physically exist, though often times it does. In this way, the readymade is *not* equivalent to a substrate or a pre-art. In practice, *physical* pre-arts and physical substrates *are* readymades.

Definition 3.7 (Primitive Pre-Art). A pre-art $p : A \rightarrow B$ is *primitive* if it cannot be expressed as a non-trivial composition of other pre-arts.

Whence, all art is made of (primitive) pre-arts.

4 The Free Group of Pre-Arts and Composition

Definition 4.1 (Composition). Given pre-arts $f : A \rightarrow B$ and $g : B \rightarrow C$, their *composition* $g \circ f : A \rightarrow C$ is the concatenation of the two transformations.

As composition is associative, and each state A admits an identity pre-art $\text{id}_A : A \rightarrow A$, so we can form the category of pre-arts.

Definition 4.2 (Category of Pre-Arts). Exercise.

Definition 4.3 (Free Group of Pre-Arts). Let $\mathcal{P}$ denote the collection of all pre-arts across all substrates. The *free group of pre-arts* $F(\mathcal{P})$ is the nonabelian free group generated by the elements of $\mathcal{P}$ under composition, with formal inverses adjoined. Each element of $F(\mathcal{P})$ is called a composition.

Remark 4.4 (Composition as Artwork). Paintings, musical pieces, installations are all intuitively compositions as a concatenation of primitive transformations. The nonabelian character of $F(\mathcal{P})$ reflects the fact that order of operations matters—the sequence of marks, gestures, and decisions is not generally commutative.

5 Types of Pre-Art

We can work in the setting of homotopy type theory (HoTT). The key feature of HoTT relevant here is that types carry internal structure: the *identity type* $a =_T b$ for two terms $a, b : T$ is itself a type, whose terms are *paths* witnessing the equivalence of a and b .

Definition 5.1 (Type of Pre-Arts). A *type of pre-arts* T is a HoTT type whose terms are elements of $F(\mathcal{P})$.

Remark 5.2. Every type sits inside $F(\mathcal{P})$, but not every subset of $F(\mathcal{P})$ constitutes a type: a type carries path structure encoding the appropriate notion of equivalence for its elements. We do not construct what makes something a type from scratch; we work in a universe where types exist and are recognized by their coherence.

Remark 5.3 (Russell’s Paradox and Universe Stratification). One might ask whether the framework encounters Russell’s paradox: if contexts are subsets of $F(\mathcal{P})$ and a context can itself be a composition in $F(\mathcal{P})$, does the “context of all contexts” generate a contradiction? It does not, for two reasons.

First, all contexts are subsets of a fixed, well-defined ambient object $F(\mathcal{P})$, constructed from concrete generators (substrates and their transformations). We do not form collections from arbitrary self-referential predicates. The “set of all sets not containing themselves” has no analogue here because membership in $F(\mathcal{P})$ is determined by the generating data, not by a free predicate.

Second, the HoTT setting stratifies the universe hierarchy explicitly: types live in universes $\mathcal{U}_0, \mathcal{U}_1, \mathcal{U}_2, \dots$, where a type in $\mathcal{U}_n$ may not contain itself as a term at the same level. If contexts-as-types live in $\mathcal{U}_0$, then the “type of all contexts” lives in $\mathcal{U}_1$, and so on. The paradox is stratified away by construction. One may reason about contexts of contexts without contradiction, provided one is attentive to universe levels.

Remark 5.4. Two compositions being equal in a type T means there is a *path* between them in T —a witness to their equivalence. What that path is depends on the type. Different types carry different notions of equivalence. This is the content of having genuine HoTT types rather than mere subsets.

Example 5.5. A series of works in six colors is a type: the terms are the six compositions, and the paths between them are the color deformations. The path structure is nontrivial and meaningful—it is what makes the series a series rather than six unrelated works.

Example 5.6. In a type organized by cruciform structure, a path between two cruciform compositions witnesses that they are cruciform *for the same structural reason*.

Definition 5.7 (Homotopy of Pre-Arts). Two pre-arts $f, g : A \rightarrow B$ are *homotopic*, written $f \sim g$, if there exists a 2-morphism $H : f \Rightarrow g$ —a path in the type they jointly inhabit, witnessing their equivalence as processes.

Theorem 5.8 (Pre-Art Category is an ∞ -Groupoid). *When all homotopy inverses are adjoined, the category of pre-arts has the structure of an ∞ -groupoid. Under the Homotopy Hypothesis, this is equivalent to a topological space: the space of all artmaking processes.*

Proof sketch. Pre-arts compose associatively, giving a category. Formal adjunction of inverses up to homotopy (Dwyer-Kan localization) produces a groupoid. The existence of homotopies between pre-arts (2-morphisms), homotopies between homotopies (3-morphisms), and so on inductively, gives the full ∞ -groupoid structure. By the Homotopy Hypothesis (Grothendieck), ∞ -groupoids model homotopy types of spaces. $\square$

$$\begin{array}{ccc} & f & \\ \emptyset_S & \begin{array}{c} \curvearrowright \\ \Downarrow H \\ \curvearrowleft \end{array} & B \\ & g & \end{array}$$

Two pre-arts f, g with the same domain and codomain, connected by a homotopy H .

A 2-morphism: equivalence of processes, not identity of results.

HoTT Flavor: Working Without Verification

In HoTT, one works with terms of a type without needing to verify in advance every property of the type. Similarly, one works with pre-art without needing to verify that a given composition is art. The judgment “this is art” happens locally, contextually, when it matters—not as a precondition for working. This is the constructive move that resolves the definitional trap.

6 The Pre-Arts Category as a Groupoid

6.1 Formal Inverses and the Groupoid Structure

We have defined the category of pre-arts, with pre-arts as morphisms and substrate states as objects. We now argue that this category admits the structure of a groupoid: every morphism has an inverse.

Definition 6.1 (Groupoid of Pre-Arts). The pre-arts category is a *groupoid* if every pre-art $f : A \rightarrow B$ admits an inverse $f^{-1} : B \rightarrow A$ such that $f^{-1} \circ f \sim \text{id}_A$ and $f \circ f^{-1} \sim \text{id}_B$, where $\sim$ denotes equivalence in the sense appropriate to the context (strict, homotopy, or parametric, as in Definitions 13.1–13.3).

Remark 6.2. The choice of equivalence relation is decisive. Under strict equivalence, most physically realized pre-arts have no inverse: one cannot unpaint a stroke, unsound a note, or un-fire a kiln. Under homotopy equivalence, inverses exist whenever there is a process—however mediated, however extended in time—that returns the substrate to a state equivalent to its initial one. Under parametric equivalence, inverses exist whenever the forgetful functor φ cannot distinguish the initial and returned states. The groupoid structure is therefore not a single structure but a family, one for each choice of equivalence.

6.2 Examples of Reversibility

The following examples develop the range of what counts as an inverse, from the trivially reversible to the structurally extreme.

Example 6.3 (Virtual Media: Color Inversion). In a digital medium, changing a substrate’s color from yellow to green is a pre-art $f : A \rightarrow B$. The inverse $f^{-1} : B \rightarrow A$ is the operation that returns the substrate to yellow. This inverse is immediately available: the undo function, the color picker, the stored state. Under strict equivalence, $f^{-1} \circ f$ is literally the identity. Digital media makes the groupoid structure maximally transparent because the state space $\mathcal{S}(S)$ is discrete and fully addressable, and any state transition is trivially reversible by direct state restoration.

More generally, the *zero operation*—blanking the canvas, resetting to the initial state $\emptyset_S$ —is a pre-art $z : A \rightarrow \emptyset_S$ whose inverse is the re-application of the composition that produced A . In digital media, this is precisely the undo stack: a record of the morphisms composing A , which can be traversed in reverse.

Example 6.4 (Physical Media: The Conceptual Undo). In physical media, strict reversal is generally unavailable. One cannot un-apply oil paint, un-fire clay, or un-cut linen. However, a weaker but conceptually meaningful inverse is available: the *conceptual undo*, in which the substrate is returned to a state that is equivalent to its initial state under an appropriate equivalence relation.

Repainting a canvas white is the paradigm case. Let $f : \emptyset_S \rightarrow A$ be the composition that produced painting A on a primed linen ground. The act of repainting white produces a morphism $g : A \rightarrow A'$, where A' is the whitened canvas. If we take parametric equivalence with forgetful functor φ erasing chromatic data below the current top layer, then $\varphi(A') \equiv \varphi(\emptyset_S)$: the whitened canvas and the blank canvas are equivalent under φ . In this sense, $g \circ f \sim_{\varphi} \text{id}_{\emptyset_S}$, and g is a φ -inverse of f .

Example 6.5 (The Extreme Case: Burning). Burning a canvas appears to be the paradigm of irreversibility. The substrate is destroyed; its chromatic, structural, and material information is dispersed as heat, light, smoke, and ash. No practical inverse exists.

And yet matter is neither created nor destroyed. The ash, the smoke, the carbon and mineral residues of the burned linen and pigment remain in the world. In principle—setting aside all practical constraints—one could: collect the ash; use it to fertilize the soil from which cotton is grown; weave new linen from that cotton; prime it with gesso; and execute the exact same composition $f : \mathcal{O}_S \rightarrow A$ on this new substrate. The result A'' is, under parametric equivalence with φ erasing provenance and material history, equivalent to A . The burn is reversed, not in time, but in the space of equivalence classes.

Burning is therefore an extreme boundary case: it maximally disperses information and maximally extends the time and effort required for inversion, but it does not in principle exit the groupoid structure. It demonstrates that the groupoid condition is a statement about equivalence classes, not about the temporal or entropic cost of achieving them.

6.3 Equivalence, Sameness, and Perception

The preceding examples make visible a general principle: the question of whether an inverse exists is not a question about physics but about the chosen equivalence relation. Different equivalence relations yield different groupoid structures over the same underlying collection of pre-arts.

Example 6.6 (Kintsugi and the Inversion of Damage). A ceramic pot is broken and repaired with gold lacquer, in the Japanese practice of *kintsugi*. Under strict equivalence, the pot before and after breakage are distinct objects; the damage is not reversed. Under parametric equivalence with φ erasing material discontinuities and retaining only overall form, the repaired pot is equivalent to the original: it has the same shape, the same function, the same identity as object. The breakage is, under φ , reversed.

Kintsugi thus naturalizes the idea that inversion can be generative—that working through the inverse of a pre-art is itself an artmaking operation, not merely a reset. The cracks become part of the composition. Reversal is a creative act.

This is the *wabi-sabi* inflection of the groupoid structure: impermanence and incompleteness are not failures of the inverse operation but features of the equivalence class to which the substrate belongs. The pot after kintsugi is equivalent to the original pot under some φ , and a new and richer composition under another. Both readings are valid; neither is privileged.

Remark 6.7 (Equivalence as Perception). The choice of φ , and hence the choice of what counts as “the same,” is not given by the mathematics but imposed by the observer, the institution, or the discourse. Sameness is perception. The mathematics does not resolve this; it parametrizes it. To each legitimate choice of perspective corresponds a legitimate groupoid structure, and hence a legitimate notion of what counts as reversal, repetition, or novelty.

6.4 The New, the Old, and the Contextual

The groupoid structure and its dependence on equivalence have a direct bearing on the question of novelty in art: what does it mean to make something new?

Remark 6.8 (The Problem of the Genuinely New). If all physical substrates are recycled—if matter on Earth cycles continuously between forms—then in what sense can any artwork be genuinely new? The question is well-posed only relative to an equivalence relation. Under sufficiently coarse φ , nothing is new: all canvases are permutations of prior canvases, all marks are rearrangements of prior marks, and the space of artmaking is a closed system traversing and re-traversing the same equivalence classes. Ends become beginnings; the new becomes old becomes new.

Under sufficiently fine φ (approaching strict equivalence), everything is new: no two substrate states are identical, and every composition produces a genuinely novel element of $F(P)$. The question of novelty does not have a context-independent answer.

Remark 6.9 (Cycles as Pre-Arts). The observation that matter and culture cycle between forms is itself an observation about the structure of $F(P)$. Consider the following recurrent patterns in the history of art and culture:

- *Stylistic cycles*: realism $\rightarrow$ abstraction $\rightarrow$ figuration $\rightarrow$ realism, recurring across Western art history from the Renaissance through Modernism, Postmodernism, and their aftermaths.
- *Economic cycles*: boom and bust, accumulation and dispersal of capital, with direct effects on what art is produced, collected, and canonized.
- *Political cycles*: eras of stability and consolidation alternating with rupture and reinvention, each producing its own characteristic art forms.
- *Material cycles*: the recycling of physical matter between substrate forms, as discussed above.

Each of these is a long-composition in $F(P)$: a sequence of pre-arts whose concatenation, viewed at the appropriate scale, approximates a return to an earlier state. They are not exact inverses—the equivalences are coarse, and the path through the cycle accrues its own composition—but they are groupoid-like in structure. The category of pre-arts, at large scale, exhibits cycles.

Proposition 6.10 (Novelty is Contextual). *What makes a composition new is not an intrinsic property of the composition but a function of the context C in which it appears and the equivalence relation $\sim_\varphi$ operative in that context. Formally: a composition $u \in F(P)$ is new relative to a context C and equivalence φ if $u \not\sim_\varphi c$ for any $c \in C$. The same u may be new relative to one context and redundant relative to another.*

Remark 6.11 (The New in Contemporary Art). What makes contemporary art contemporary is not its material composition—which is always recycled from prior materials—but its *context*: the institutional frame, the discourse, the historical moment, the exhibition, the lineage it cites and departs from. A monochrome painted today is not the same object as a monochrome painted in 1950, even if they are parametrically equivalent under coarse φ , because the context C in which the contemporary monochrome appears is different, and the morphisms relating it to that context are different. Context is the locus of novelty.

This formalizes Duchamp’s insight that the readymade is constituted not by its material properties but by its placement in a context—a gallery, a title, a date, a signature. The readymade is an element of $F(P)$ whose novelty is entirely contextual. The groupoid structure makes this precise: the readymade and its source object are strictly equivalent as physical configurations, and contextually non-equivalent as elements of $F(P)$ embedded in distinct contexts.

7 Contexts

Definition 7.1 (Context). A *context* is a subset $\mathcal{C} \subseteq F(\mathcal{P})$.

That is, a context is constituted by its pre-arts.

Example 7.2. The walls of a gallery, the lighting, the spatial configuration, the labels are all pre-arts and all contribute to their shared context.

Definition 7.3 (Well-Formed Context). A context $\mathcal{C}$ is *well-formed* if it carries the structure of a HoTT type.

Remark 7.4. Not every subset of $F(\mathcal{P})$ need be well-formed. A well-formed context is one wherein its elements admit a coherent notion of identity (path structure) encoding when two compositions in $\mathcal{C}$ are equivalent in the relevant sense.

Because a general context $\mathcal{C}$ is merely a subset, it is not closed under composition, motivating the following definition:

Given a context $\mathcal{C} \subseteq F(\mathcal{P})$, we define the *subgroup generated by $\mathcal{C}$* :

$$\langle \mathcal{C} \rangle := \left\{ \text{all finite products } c_1^{\epsilon_1} \cdots c_n^{\epsilon_n} \mid c_i \in \mathcal{C}, \epsilon_i = \pm 1 \right\} \leq F(\mathcal{P}).$$

This subgroup is not necessarily free on the set $\mathcal{C}$, as relations may exist among its elements.

To recover freeness, we pass to the primitives:

Definition 7.5 (Primitive Support). The *primitive support* of a context $\mathcal{C}$, denoted $\mathcal{P}_{\mathcal{C}} \subseteq \mathcal{P}$, is the set of all primitive pre-arts that appear in the decompositions of elements of $\mathcal{C}$.

Then $F(\mathcal{P}_{\mathcal{C}})$ is free on $\mathcal{P}_{\mathcal{C}}$ and embeds as a free subgroup of $F(\mathcal{P})$.

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\iota} & \langle \mathcal{C} \rangle \\ \downarrow \iota & \swarrow \iota & \downarrow \iota \\ F(\mathcal{P}_{\mathcal{C}}) & \xrightarrow{\iota} & F(\mathcal{P}) \end{array}$$

Definition 7.6 (Free Group on a Context). Let $\mathcal{C} \subseteq F(\mathcal{P})$ be a context. By abuse of language, we define the *free group generated by $\mathcal{C}$* , denoted $F(\mathcal{C})$ as $F(\mathcal{P}_{\mathcal{C}})$, so that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\iota_{\mathcal{C}}} & F(\mathcal{C}) \\ \downarrow \iota_{\mathcal{P}} & \swarrow \exists \bar{\iota} & \\ F(\mathcal{P}) & & \end{array}$$

Here $\bar{\iota} : F(\mathcal{C}) \rightarrow F(\mathcal{P})$ is the unique group homomorphism sending each generator $c \in \mathcal{C}$ (viewed as a generator of $F(\mathcal{C})$) to the corresponding element $c \in F(\mathcal{P})$. This map encodes the internal structure of each element of $\mathcal{C}$ as a word in the original primitives $\mathcal{P}$.

A natural question is whether the quotient group $F(\mathcal{P})/F(\mathcal{C})$ exists. The answer is, theoretically yes, if and only if $F(\mathcal{C})$ is a normal subgroup of $F(\mathcal{P})$. In physical practice, most pre-arts are irreversible (entropy-increasing).

Remark 7.7. An exhibition is a context in which the substrates include a physical or virtual space: a gallery, an apartment, a digital environment, and the pre-arts that transform the space. Whether a given exhibition is well-formed is a question the theory allows one to ask, but does not require one to answer in advance.

Remark 7.8. A series is a context. A single wall is a context. A gallery show is a context. An artist's entire body of work is a context. A well-curated series is likely well-formed; an arbitrary group show may not be. The theory provides the vocabulary to make this distinction precise without mandating that all exhibitions meet the stronger condition.

Remark 7.9 (Exhibition as Singleton and as Subset). An exhibition admits two compatible readings within $F(\mathcal{P})$. In the first, the exhibition is a *single element*: the composition $f = p_n \circ \cdots \circ p_1 \in F(\mathcal{P})$ obtained by concatenating all constituent pre-arts in sequence. In this reading, the show is one morphism on the substrate of the space. In the second, the exhibition is a *subset*: $\{p_1, \dots, p_n\} \subset F(\mathcal{P})$, the collection of constituent pre-arts considered individually. Both readings are valid and complementary. The first emphasizes the exhibition as a unified composition; the second emphasizes its decomposition into parts.

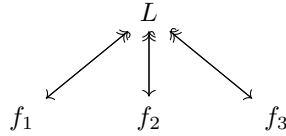
The decomposition $p_n \circ \cdots \circ p_1$ is itself a *string*—a word in the free group $F(\mathcal{P})$. In a free group, there are no relations: distinct strings are genuinely distinct elements. This means two exhibitions that result in visually identical configurations but were produced by different sequences of decisions are formally different elements of $F(\mathcal{P})$. They may be homotopic—connected by a 2-morphism witnessing the equivalence of their processes—but they are not identical. The difference between them is style at the level of exhibition-making.

A commutative diagram displaying the decomposition of a composition is itself a string: a pre-art on the substrate of mathematical notation, an element of $F(\mathcal{P})$ in its own right. The meta-level description of a composition is itself a composition. The structure is self-similar.

Proposition 7.10 (The Gallery is Pre-Art). *The physical and institutional infrastructure of an exhibition—walls, lighting, labels, the institution itself—are pre-arts: morphisms on the substrate of the space. To exhibit is to compose one’s pre-arts with the pre-arts of the space. This composition is generally noncommutative: the order in which decisions are made (what goes where, what is adjacent, what is lit how) affects the result.*

Example 7.11. Sarah Sze’s installations make the full morphism visible. Rather than presenting only the codomain (the finished work), Sze includes the depleted paint tubes, scattered materials, and detritus of process. She does not hide the pre-arts that constitute the work; she exhibits them alongside their results. Her practice is a direct instantiation of the claim that the infrastructure of making is pre-art too.

Definition 7.12 (Local Coherence). A context is *locally coherent* at a sub-context $\mathcal{C}' \subset \mathcal{C}$ if the terms of $\mathcal{C}'$ share a common type: there exists a type T such that every composition in $\mathcal{C}'$ is a term of T . Informally, the works in a locally coherent sub-context are all “about” the same thing, or proceed by the same structural principle.



A locally coherent wall: compositions f_1, f_2, f_3 sharing a common limit L .

L injects into each work (it is present in each) and each work projects onto L (it is extractable from each).

Remark 7.13. Large group exhibitions are typically only locally coherent: there exist sub-contexts with no shared type, walls with no common limit. This is not a failure of curation; it is an honest structural description of how large shows are organized. Global coherence across an entire group show is rare and not required.

8 Design Principles, Universal Pre-Art and Accessibility

Definition 8.1 (Design Principle). A *design principle* is a context constructed by collecting together all compositions in $F(\mathcal{P})$ that exhibit a given structural property.

Remark 8.2. We do not build the principle from axioms; we gather everything that already has the property and declare that the context. This is analogous in spirit to the Dwyer-Kan localization: the structure is not imposed but allowed to emerge from the collection.

Example 8.3. Examples of design principles include

- **Cruciform composition:** the primary visual axis forms a cross.
- **Repetition:** a primitive pre-art or motif appears multiple times with controlled variation.

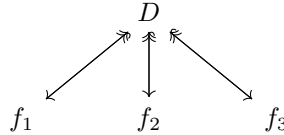
- **Contrast:** adjacent substrate states exhibit maximal difference along some perceptual axis.
- **Symmetry:** the composition is invariant under a group action.
- **Golden ratio / rule of thirds:** compositional proportions satisfy given numerical relationships.

Remark 8.4. Whether a design principle, so constructed, is additionally a well-formed HoTT type depends on whether the collection carries coherent path structure. For some principles (symmetry under a group action, color variation) this is plausible; for others (contrast, rule of thirds) it is less clear. We do not require design principles to be HoTT types.

Remark 8.5 (Design Principles as Cones). The appropriate categorical picture for a design principle is not equality of elements but a *cone*. Let D denote the apex—the design principle itself, considered as an abstract object. A design principle forms a cone if:

- every composition f in the collection maps to D (the principle is extractable from each work), and
- D maps back into each f (the principle is present in each work).

This is precisely the limit/colimit structure: D is simultaneously a pullback (extracted from the works) and a pushforward (injected into them). Notably, this is the same structure as the locally coherent wall diagram in Section 7: a design principle *is* a locally coherent context, with the design principle apex playing the role of L . The two notions are unified by the cone.



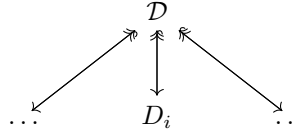
A design principle as a cone: compositions f_1, f_2, f_3 sharing apex D .

Each work maps to D (the principle is extractable); D maps into each work (the principle is present).

No equality between f_1, f_2, f_3 is required or asserted.

Remark 8.6. If two design principles, constructed by this collection process, turn out to gather the same compositions, they are the same context and hence the same principle, regardless of their labels. We do not need to prove this in any given case; if two labels collect the same elements, they are interchangeable. Elements of a design principle are not required to be mutually compatible in any stronger sense—the cone condition is sufficient.

Remark 8.7. We can also construct the context of all design principles, $\mathcal{D}$, by collecting all design principles $\{D_i\}_i$ together, having them all point to $\mathcal{D}$, again forming a cone.



Definition 8.8 (Universal Pre-Art for a Context). Let $\mathcal{C} \subseteq F(\mathcal{P})$ be a context. A composition $u \in \mathcal{C}$ is a *universal pre-art* for $\mathcal{C}$ if for every $c \in \mathcal{C}$ there exists a unique set-theoretic function $u \rightarrow c$.

That is, u is an initial object of $\mathcal{C}$ viewed as a discrete category, with morphisms being set-theoretic functions between elements of $F(\mathcal{P})$.

Remark 8.9 (Morphisms as Set-Theoretic Functions). Since $\mathcal{C} \subset F(\mathcal{P})$, the morphisms $u \rightarrow c$ in the universal pre-art definition are set-theoretic functions as functors between discrete categories, enabling the full vocabulary of category theory. When $\mathcal{C}$ happens to be a well-formed context or a HoTT type, richer morphism structure becomes available.

Remark 8.10. Since all (visual) art that is generally accepted as “great” has design principles, it is natural that all universal pre-arts exhibit design principles. Namely, a pre-art is universal if it is an arrow in any design principle cone.

Definition 8.11 (Universally Principled Context). A context $\mathcal{C} \subseteq F(\mathcal{P})$ is *universally principled*—henceforth a *universal context* (by abuse of language)—if there exists at least one set-theoretic function $\phi : \mathcal{C} \rightarrow \mathcal{D}$ such that for every $f \in \mathcal{C}$, $\phi(f)$ lands in at least one fiber $D_i \subseteq \mathcal{D}$.

That is, every element of $\mathcal{C}$ exhibits at least one design principle.

Remark 8.12 (On the Non-Uniqueness of ϕ). The map $\phi : \mathcal{C} \rightarrow \mathcal{D}$ witnessing universal principledness is in general far from unique. A single composition $f \in \mathcal{C}$ may exhibit multiple design principles simultaneously. The condition requires only existence: some ϕ witnesses that every element of $\mathcal{C}$ has at least one design principle. The multiplicity of ϕ reflects the interpretive richness of works.

Definition 8.13 (Universal Expressionism). *Universal expressionism* is defined by (pre)arts existing in universal contexts.

Remark 8.14 (The universally principled condition is empirical, not mathematical). The definition of a universal pre-art for a context $\mathcal{C}$ requires no reference to design principles; compatibility with $\mathcal{D}$ is a separate, weaker condition imposed afterward. Insofar as there is broad agreement that recognized great art exhibits design principles, requiring $\mathcal{C}$ to be universally principled is a reasonable practical constraint.

Theorem 8.15 (Participation via Adjunction). *Let $\mathcal{C}$ be a universal context, and let $u \notin \mathcal{C}$ be a composition such that:*

1. *there exists a unique set-theoretic function $u \rightarrow c$ in $F(\mathcal{P})$ for every $c \in \mathcal{C}$, and*
2. *there exists at least one $\phi' : \mathcal{C} \sqcup \{u\} \rightarrow \mathcal{D}$ extending some ϕ on $\mathcal{C}$.*

Then the adjoined context $\mathcal{C} \sqcup \{u\}$ is again universal, u is a universal pre-art for $\mathcal{C} \sqcup \{u\}$, and u is a valid contributor to the context.

Proof. Condition (1) gives u the initial object property in $\mathcal{C} \sqcup \{u\}$: it maps uniquely to every existing element of $\mathcal{C}$, and the identity serves as the unique morphism $u \rightarrow u$. Condition (2) ensures the extended context remains universally principled: ϕ' witnesses that u itself exhibits at least one design principle, and restricts to a valid ϕ on $\mathcal{C}$. $\square$

Remark 8.16 (Iconography as the Entry Point). The simplest compositions satisfying condition (2) of Theorem 8.15 are iconographic forms. The adjunction $\mathcal{C} \sqcup \{u\}$ makes the participation step explicit: the new pre-art is not merely compatible with the context, it is adjoined to it, enlarging the context while preserving its universal principledness.

Accessibility: The Mathematical Statement

The constraint of participation is compatibility through adjunction into a universally principled context. Anybody can draw simple shapes, hence anyone can produce iconographic forms—compositions of elementary pre-arts (lines, marks, shapes) that exhibit design principles. Since iconographic forms satisfy condition (2), and since any participant capable of marking a substrate can produce them, participation is made radically accessible under Universal Expressionism.

9 Contexts as Types: Pre-Arts in Homotopy Type Theory

9.1 Every Pre-Art Exists in a Context

The groupoid structure of the pre-arts category established in the preceding section gives us direct access to the machinery of Homotopy Type Theory. The key move is the identification of contexts

with types: a well-formed context $C \subseteq F(P)$ is precisely a HoTT type, and the morphisms between its elements—the paths, homotopies, and higher homotopies—encode the equivalences relevant to that context.

This has an immediate consequence for the status of pre-arts: every pre-art exists in some context.

Definition 9.1 (Contextual Existence). Every pre-art $f \in F(P)$ is a term of at least one well-formed context $C \subseteq F(P)$. That is, every pre-art has a type.

Remark 9.2. This axiom is not a strong constraint. The minimal context containing f is the singleton $\{f\}$, which carries trivial type structure (the only path is id_f). But even this trivial context places f within the HoTT framework and makes its identity structure available for analysis. The axiom asserts that no pre-art floats free of all context: every morphism between substrate states exists relative to some structured collection.

9.2 Readymades and the Context-Dependence of Interpretation

The Contextual Existence Axiom is the formal statement of Duchamp’s insight. A readymade $r = \text{id}_{(\emptyset_X, \emptyset_I)} : (\emptyset_X, \emptyset_I) \rightarrow (\emptyset_X, \emptyset_I)$ is a pre-art. As a pre-art, it is a term of some context C . Different contexts C_1, C_2 containing r yield different types, and the same physical object r —the same urinal, the same snow shovel, the same bottle rack—inhabits different types depending on the context.

Example 9.3. Let r denote Duchamp’s *Fountain* (1917), considered as a pre-art: the identity morphism on the urinal-as-substrate. The context $C_1 = \{\text{plumbing fixtures in a hardware store}\}$ gives r a type in which the relevant paths are functional (does it carry water? does it seal?), and the design principles operative in C_1 are engineering ones. The context $C_2 = \{\text{works submitted to the Society of Independent Artists, 1917}\}$ gives r a type in which the relevant paths are institutional and art-historical, and the design principles operative in C_2 include the readymade gesture, institutional critique, and the question of what counts as art. Under C_1 , r is a plumbing fixture. Under C_2 , r is one of the most significant artworks of the twentieth century. The object $r \in F(P)$ is the same element; its type, and hence its interpretation, is determined by the context.

Remark 9.4 (Context Changes Interpretation, Not Identity). The dependence of interpretation on context does not threaten the identity of pre-arts as elements of $F(P)$. The free group $F(P)$ is a fixed ambient structure; contexts are subsets of it. The same element $r \in F(P)$ is a term of multiple types simultaneously. What changes between contexts is not r itself but the path structure available to it—the equivalences, the homotopies, the design principles that apply. This is the content of type-membership in HoTT: a term may inhabit multiple types, and its properties in each type are determined by that type’s path structure.

9.3 The Groupoid and HoTT Access

The groupoid structure of the pre-arts category (Section [NEW GROUPOID SECTION]) is precisely what gives HoTT its purchase on pre-arts. In HoTT, the identity type $f =_C g$ for two terms $f, g : C$ is itself a type, whose terms are *paths* from f to g in C . The invertibility of paths in HoTT—the fact that every path $p : f =_C g$ has an inverse path $p^{-1} : g =_C f$ —corresponds directly to the groupoid structure of the pre-arts category.

Proposition 9.5 (HoTT Access via Groupoid Structure). *A well-formed context $C \subseteq F(P)$ is a HoTT type if and only if its path structure is groupoid-like: composition of paths is associative (up to higher homotopy), identity paths exist, and every path is invertible. The groupoid of pre-arts, established by the equivalence-relative inverse construction of Section [NEW GROUPOID SECTION], provides exactly this structure.*

Remark 9.6 (Types and the Truncation Hierarchy). Not all well-formed contexts have the same homotopy complexity. A context C is a:

- *(−1)-type (mere proposition)*: at most one path between any two elements; the context asserts membership without structure.
- *0-type (set)*: all paths between elements are equal; the context carries no homotopy information above dimension 0.
- *1-type (groupoid)*: paths form a groupoid; distinct paths between the same elements are possible and meaningful. This is the natural level for contexts organized by design principles, where two paths $H, K : f \sim_C g$ may witness the same equivalence by different homotopies.
- *∞-type*: full higher homotopy structure. This is the level of the full pre-arts ∞-groupoid of Theorem 5.8.

Most artistically interesting contexts live at the 1-type level: they are groupoids, with non-trivial but not infinitely complex homotopy structure. The truncation level of a context is a measure of its interpretive complexity.

Remark 9.7 (The New as a Type-Theoretic Statement). From the HoTT perspective, a composition u is new relative to a context C if u is not pathwise connected to any existing element of C : there is no path $p : u =_C c$ for any $c \in C$. The context $C \sqcup \{u\}$ is the adjoined context, and Theorem 7.15 (Participation via Adjunction) ensures that if u satisfies the universally principled conditions, the adjoined context remains well-formed. Adding a new element to a context is a well-typed operation, and the novelty of u is witnessed by the absence of paths from u to prior elements—an intrinsically contextual statement.

10 Ideas

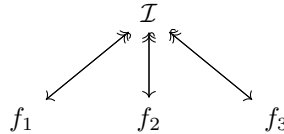
The framework as developed operates on material substrates: canvas, wall, silence, space. Yet artistic practice is inseparable from ideas—concepts, intentions, references, lineages. We now extend the framework to accommodate ideas formally, without disturbing the existing structure.

Definition 10.1 (Idea). An *idea* is the cone over all pre-arts that instantiate, represent, or embody it—collected via the Dwyer-Kan localization of all compositions containing it as a component. Formally, let $\mathcal{I}$ denote an idea considered as an apex object. The idea $\mathcal{I}$ is constituted by the cone:

$$\mathcal{I} \hookrightarrow f_i, \quad f_i \twoheadrightarrow \mathcal{I}$$

for all $f_i \in F(\mathcal{P})$ that instantiate $\mathcal{I}$: the idea injects into each instantiation (it is present in each work) and is extractable from each (it survives into any representation of it).

That is,



Remark 10.2 (Naturality of the Localization). Ideas tend toward physical instantiation. If an idea has sufficient cultural pressure, it gets made—by one artist, or independently by several. The cone of an idea fills from below, driven not by decree but by the iterative search of practitioners traversing $F(\mathcal{P})$. The Dwyer-Kan style collection is therefore not an artificial imposition: it describes what actually happens. The idea of mechanical reproduction, the idea of the monochrome, the idea of institutional critique—each was instantiated multiply and independently. The cone merely formalizes the inevitability.

Example 10.3 (Design Principles as a Special Case of Ideas). Design principles (Definition 7.1) are ideas in the sense of Definition 10.1: they are cones over all compositions exhibiting a given structural property. The apex D of a design principle cone is the idea of that principle—cruciformity, symmetry, repetition—and the cone condition is identical. Ideas are thus strictly more general: every design principle is an idea, but not every idea is a design principle. Design principles are the *formal-structural* subclass of ideas; they carry an additional guarantee that the property is perceptually or compositionally checkable. Ideas lacking this checkability—grief, duality, critique—are semantically rich but do not by themselves guarantee that a work will cohere visually or formally. This is precisely why universal principledness is defined via design principles rather than ideas in general: an artist may work from pure intuition, arriving at an idea only retrospectively, and the work is still art and still exhibits design principles. The design principle is the formal guarantee; the idea is the semantic depth. Both may be present; neither entails the other.

Remark 10.4 (Ideas as Product Substrates). A work that operates simultaneously on a material substrate S_1 and a conceptual substrate S_2 acts on the product substrate $S_1 \times S_2$. A pre-art $f : A \rightarrow B$ with $A, B \in \mathcal{S}(S_1 \times S_2)$ transforms both the material state and the conceptual state simultaneously. The initial state is $(\emptyset_{S_1}, \emptyset_{S_2})$: blank canvas and latent concept. The product construction requires no new machinery; it is a standard operation on the substrates already present in the framework.

Remark 10.5 (Works Refer to Prior Works and Ideas). A composition that instantiates an idea $\mathcal{I}$ thereby points to the entire cone of $\mathcal{I}$: every past, present, and future instantiation, every work that has previously populated the fiber. A new work on the theme of entropy does not encounter entropy as a blank concept; it encounters entropy as already mediated by the second law, by a century of thermodynamic imagery. The idea-arrow in the composition is not a pointer to an abstract Platonic form but to a populated cone, a live context. No idea is wholly new; every idea-arrow carries its lineage as part of its domain. Naturally, we can always project down from the idea cone to restrict to past instantiations only, and then adjoin a new instantiation, to make time explicit. Notwithstanding, ideas themselves transcend time.

Remark 10.6 (Well-Definedness and Quotients). The inclusion of idea-arrows in compositions does not in general compromise well-definedness. Two formally distinct compositions in $F(\mathcal{P})$ that instantiate the same idea via different sequences of idea-arrows are generally different elements of $F(\mathcal{P})$ —different processes, different paths through the iterative search—and may be homotopic (connected by a 2-morphism) without being identical. If, however, a particular context requires that two compositions be identified—for instance, because one is interested only in the idea instantiated and not the material path taken—one may pass to the quotient $F(\mathcal{P})/\sim$ under the appropriate equivalence relation $\sim$.

Remark 10.7 (On Temporal Order and the Arrow of Composition). The nonabelian structure of $F(\mathcal{P})$ encodes that order matters. It is natural to read this order as chronological: first this mark, then that one. But the framework makes no explicit reference to time. The ordering in a composition $p_n \circ \cdots \circ p_1$ is formal, not temporal.

This becomes significant when ideas are included. An artist may work intuitively—no declared idea, pure material process—and arrive at an idea only retrospectively, after the work is complete. Under a strictly chronological reading, the idea-arrow cannot appear “before” the work because it did not precede it in time. But under the formal reading, there is no obstacle: the idea-arrow can be placed anywhere in the composition that makes the cone structure coherent. The “initial condition” $\emptyset_S$ is a formal choice, not a claim about what happened first.

One may go further. A homotopy $H : f \Rightarrow g$ is a path parametrized by $[0, 1]$; reversing the parametrization $t \mapsto 1 - t$ reverses the direction, exchanging domain and codomain. There is no preferred orientation. The arrow of composition is a convention.

Whether this implies that idea-arrows can coherently run backward in time—the idea being in some sense caused by the work that instantiates it, rather than preceding it—is an open question the framework permits but does not resolve. We leave it as an invitation.

11 Conversation Between Works and Between Exhibitions

Definition 11.1 (Morphism Between Compositions). A *morphism between compositions* $\phi : f \rightarrow g$ in a type T is a path in T from f to g : a witness that f and g are related within the type. Such morphisms encode relationships of influence, response, variation, critique, or homage—any structural connection that places two works in the same type and witnesses their relationship.

Definition 11.2 (Functor Between Contexts). A *functor* $\Phi : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ between two contexts is a map sending compositions in $\mathcal{C}_1$ to compositions in $\mathcal{C}_2$, preserving composition and identity. A functor between exhibitions is what it means for two exhibitions to be “in conversation.”

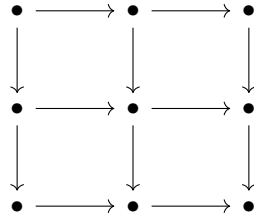
Example 11.3. The historical Color Field painters (Frankenthaler, Rothko, Kelly) and a contemporary digital color field practice constitute two contexts connected by a functor: shared pre-arts (the investigation of color as primary substrate transformation) map across the historical gap, with the contemporary practice transforming the historical compositions through a new substrate. The functor is the lineage.

12 The Iterative Search

Definition 12.1 (Iterative Search). *Iterative search* is the traversal of the graph of pre-art relationships in $F(\mathcal{P})$ —composing, testing type membership, following morphisms between types—in pursuit of compositions that cohere into a locally coherent context.

The artmaking process, under this framework, is a search through the space of compositions. One begins with a primitive pre-art or a found composition, composes further, tests whether the result inhabits a desired type, revises, and composes again. The process terminates locally when sufficient coherence is achieved.

Remark 12.2. The commutative diagram as an art object is self-referential in precisely this sense: it depicts the categorical structure used to analyze it. It is a composition whose characteristic type is “depicts morphisms between substrate states,” and whose composition with this framework constitutes a fixed point. The diagram is simultaneously an element of $F(\mathcal{P})$ and an illustration of the structure of $F(\mathcal{P})$.



A commutative diagram as art object: symmetric under $\mathbb{Z}/4\mathbb{Z}$, validly hung in four orientations.

Bullets replace labels to foreground pure structure. The reversal of all arrows is a distinct but equally valid composition. The design principle is the diagram itself, prior to any mathematical content.

Remark 12.3 (The Paper as Exhibition). This paper is itself an element of $F(\mathcal{P})$: a composition of pre-arts (definitions, propositions, diagrams, printed text, spatial arrangement on a page or wall) on the substrate of mathematical writing. It may be exhibited as a purely conceptual show by printing its definitions, propositions, and diagrams directly onto gallery walls, with no other objects present. In that instantiation, the paper becomes an exhibition that is an element of the type it describes—a fixed point. The theory applies to itself. This is not a merely clever observation; it is in direct lineage from Kosuth’s *Art as Idea as Idea* (1966), but with an explicit formal structure beneath the gesture rather than the gesture alone. Whether such an exhibition is well-formed in the sense of Definition 6.3 is itself an open and interesting question.

13 Ideas as Cones: Geometry and Symmetry in the Pre-Arts Framework

13.1 Introduction

In the Pre-Arts framework, artworks and pre-arts are understood as morphisms between substrate states. A natural question arises: what is the categorical structure of *ideas* themselves? We argue that ideas admit a precise mathematical characterization as objects equipped with both limit and colimit cone structure over diagrams of pre-arts. This chapter develops that characterization, identifies a necessary axiom, and draws out its implications for the geometry and symmetry of ideas, the duality of observation and expression, and the natural inclusivity of the framework.

13.2 Background: Limits, Colimits, and Cones

Let $\mathcal{C}$ be a category. Recall that given a diagram $F : \mathcal{J} \rightarrow \mathcal{C}$ indexed by a small category $\mathcal{J}$, a *cone* over F with apex N consists of an object $N \in \mathcal{C}$ together with a family of morphisms

$$\psi_j : N \rightarrow F(j), \quad j \in \mathcal{J},$$

such that for every morphism $f : j \rightarrow j'$ in $\mathcal{J}$, the diagram

$$\begin{array}{ccc} N & \xrightarrow{\psi_j} & F(j) \\ & \searrow \psi_{j'} & \downarrow F(f) \\ & & F(j') \end{array}$$

commutes. The *limit* $\varprojlim F$ is the universal such cone: the apex through which every other cone factors uniquely.

Dually, a *cocone* under F with nadir N consists of morphisms

$$\phi_j : F(j) \rightarrow N$$

commuting with all $F(f)$. The *colimit* $\varinjlim F$ is the universal cocone.

When working in a homotopy-theoretic setting, we replace these with homotopy limits and colimits in the sense of Dwyer–Kan, localizing at weak equivalences so that essentially equivalent instantiations are identified.

13.3 The Pointability Axiom

Before characterizing ideas categorically, we distinguish between two classes of ideas within the framework.

Definition 13.1 (Well-Formed Idea). An idea I is *well-formed* if it admits a precise symbolic expression—mathematical, logical, or formal—such that the cone structure of Definition 12.3 is coherent: the limit and colimit over the diagram of instantiations exist and are consistent. Well-formed ideas are those whose internal structure is sufficiently logical to support the full categorical apparatus.

Remark 13.2. Mathematical proofs are the paradigm of well-formed ideas. A proof of a theorem is an idea with natural categorical geometry: the theorem is the apex of the limit cone (the statement shared by all valid proofs), and each proof is an instantiation in the colimit cone (a particular path to the theorem). The idea of *entropy*, the idea of *cruciform composition*, and the idea of *accumulation* are well-formed in this sense: they admit formal characterization sufficient for the cone structure to be coherent and for the design principle condition to be checkable.

Definition 13.3 (Non-Well-Formed Idea). An idea I is *non-well-formed* if it resists full symbolic expression: it cannot be captured by any formal system without remainder. Grief, the sublime, the inarticulate, the pre-linguistic—these are ideas that participate in the framework only through the pointing map, and whose cone structure is thin or degenerate.

Remark 13.4 (Mathematical Honesty). This distinction is a statement of mathematical honesty. Gödel’s incompleteness theorems establish that no sufficiently expressive formal system can prove all truths about itself. Tarski’s undefinability theorem establishes that truth is not formally definable within the system that expresses it. The pre-arts framework, insofar as it employs formal mathematics, inherits these limitations. There are ideas—artistically valid, humanly significant, practically potent—that the formalism cannot fully capture. The distinction between well-formed and non-well-formed ideas is the framework’s acknowledgment of its own boundary.

This is not a weakness. A theory that claims to formalize everything is either wrong or trivial. The pre-arts framework makes a rich and precise contribution to the theory of well-formed ideas and remains honest about the boundary.

Remark 13.5 (Symbols as Pointing Maps for Non-Well-Formed Ideas). Non-well-formed ideas are not excluded from the framework. They enter through symbolic pointing: a mark, a word, a gesture that says “this” without fully articulating what “this” is. Every pre-art that instantiates grief, or the sublime, or an inarticulate intuition is a pointing map from a symbolic expression to the non-well-formed idea. The idea is in the diagram; its cone is thin; its geometry is opaque. Mathematics can gesture at it but not analyze it fully. This is the correct and honest use of the formalism.

Remark 13.6 (Non-Well-Formed Ideas and Collapse). Ideas that resist expression—that are not pointed at by any pre-art, that generate no instantiations, that are not taken up in any context—collapse to zero: they do not persist in the framework. Non-well-formed ideas with thin cone structure may or may not persist, depending on whether the pointing maps directed at them are culturally sustained. This is consistent with the observation in Remark 12.8: cone density governs persistence. Non-well-formed ideas with no cone structure at all are, from the framework’s perspective, absent.

The Pointability Axiom is now restricted to well-formed ideas.

Definition 13.7 (Pointability Axiom, Revised). Every *well-formed* idea admits at least one expression. That is, for any well-formed idea I , there exists at least one pre-art P and a morphism $P \rightarrow I$ or $I \rightarrow P$ in the relevant diagram. Moreover, well-formed ideas exist as terms of the pre-arts type: they are elements of some well-formed context $C \subseteq F(P)$, and their cone structure is coherent with the path structure of C .

Remark 13.8. The restriction to well-formed ideas is necessary for the Pointability Axiom to be a formal claim rather than an aspiration. For non-well-formed ideas, we cannot guarantee that a coherent expression exists; we can only guarantee that pointing maps may be directed at them. The axiom, so restricted, remains weak but formally precise.

Remark 13.9 (Well-Formed Ideas and Physical Instantiation). Well-formed ideas—those with sufficient logical structure to support coherent cone geometry—tend toward physical instantiation. An idea with a dense, coherent colimit cone generates many expressions across many media; it is the kind of idea that multiple artists, working independently, arrive at in a given historical period. The cone of *mechanical reproduction*, the cone of the *monochrome*, the cone of *institutional critique*: each was populated independently and multiply. This is not coincidental. Well-formed ideas have an internal pressure toward expression, because their structure makes them legible, transmissible, and generative. Their colimit cones fill from below, driven by practitioners traversing $F(P)$.

Non-well-formed ideas, by contrast, may resist this pressure. They can be pointed at but not fully transmitted. They generate thin colimit cones, sporadic instantiations, and often depend on a single practitioner to sustain them. When that practitioner’s work falls out of context, the idea may not persist.

13.4 Ideas as Bidirectional Cone Structures

Definition 13.10 (Idea). Let $\mathcal{P}$ denote the category of pre-arts. An *idea* I is an object equipped with:

1. A diagram $F : \mathcal{J} \rightarrow \mathcal{P}$ of pre-arts indexed by a small category $\mathcal{J}$,
2. A limit cone structure: a universal cone $\varprojlim F \simeq I$ with morphisms from expressions into I ,
3. A colimit cone structure: a universal cocone $\varinjlim F \simeq I$ with morphisms from I into expressions.

We do not assert that the limit and colimit coincide or are equivalent. Rather, I is the object that admits both universal properties simultaneously over the diagram of its expressions.

The bidirectionality is not a defect to be resolved but the essential content of the definition. It reflects a genuine duality in creative practice:

Proposition 13.11 (Creative Duality). *The two cone orientations correspond to two directions of creative process:*

$$\begin{aligned} \text{Limit direction:} \quad & \text{expressions} \rightarrow I \quad (\text{idea crystallizes from work}), \\ \text{Colimit direction:} \quad & I \rightarrow \text{expressions} \quad (\text{work instantiates from idea}). \end{aligned}$$

Neither direction is privileged. Both are valid and both occur in practice.

This dissolves the apparent conflict between concept-led and process-led art-making. The question “does the idea precede the work, or does the work generate the idea?” becomes a question of which cone one is tracing—not a deep metaphysical dispute about creative priority.

The bidirectional cone structure of a well-formed idea I (Definition 12.3) yields two distinguished objects: the limit and the colimit over the diagram of instantiations.

Remark 13.12 (The Limit as Minimal Element). The limit $\varprojlim F \simeq I$ is the *crux* of the idea: the minimal shared content present in every instantiation. It is the intersection of all expressions, the common core that any instantiation of I must contain. For the idea of *cruciform composition*, the limit is the abstract cross—the shared structural property that every cruciform composition instantiates. For the idea of *entropy*, the limit is the formal property of irreversible dispersal, present in every work that instantiates entropy regardless of medium, material, or cultural context.

The limit is minimal in the sense that it is the largest object that maps into every instantiation: it is “what all instantiations have in common,” made precise.

Remark 13.13 (The Colimit as Maximal Element). The colimit $\varinjlim F \simeq I$ is the *total idea*: the object generated by collecting all instantiations together, with their relationships. It is the largest object generated by all expressions: the idea as fully elaborated by every work that has ever instantiated it, across all media and all historical moments. For the idea of *entropy*, the colimit includes the second law of thermodynamics, a century of thermodynamic imagery, Robert Smithson’s Partially Buried Woodshed, digital data corruption art, and every future instantiation yet to be made.

The colimit is maximal in the sense that every instantiation maps into it: it is “everything the idea has become and will become,” made precise.

Remark 13.14 (Limit and Colimit as Creative Coordinates). The limit and colimit of a well-formed idea give the artist two conceptual coordinates:

- The *limit* tells the artist what is necessary: the minimal content that any valid instantiation of this idea must contain. To ignore the limit is to produce a work that, under the operative equivalence, does not instantiate the idea at all.
- The *colimit* tells the artist what has been accumulated: the full cone of prior instantiations, the lineage the new work joins. A new instantiation is adjoined to the colimit; it must map into the colimit without being identified with any prior element (unless the equivalence collapses them).

This replaces vague talk of “being in dialogue with” or “responding to” prior work with a precise categorical statement: a new work instantiating idea I is an element of the colimit cone, and its relationship to prior works is encoded in the morphisms of that cone.

Remark 13.15 (Non-Well-Formed Ideas Have No Guaranteed Limit or Colimit). For non-well-formed ideas, the limit and colimit may not exist, may not be coherent, or may depend on the choice of diagram in ways that cannot be made canonical. This is precisely the content of their non-well-formedness: they resist the dual universal property. Mathematics can point at them; it cannot analyze their structure with the same precision. This is an accurate description of the difficulty of working with subliminal, inarticulate, or pre-linguistic ideas in any formal system, and the pre-arts framework makes no stronger claim about them.

13.5 Geometry and Symmetry of Ideas

The cone structure endows ideas with a genuine geometry in the categorical sense: a shape, a direction, universal properties, and a relationship to a diagram of expressions.

Remark 13.16 (Geometric Content). An idea is not merely a fuzzy semantic object or a psychological state. It is an object with:

- *Shape*: determined by the index category $\mathcal{J}$,
- *Direction*: incoming (limit) and outgoing (colimit) morphisms,
- *Universal property*: every compatible cone or cocone factors uniquely through I ,
- *Local geometry*: the incoming and outgoing directions together give I something analogous to a tangent structure at a point.

The symmetry of ideas follows from the self-dual structure of the definition. Under categorical duality $\mathcal{P} \mapsto \mathcal{P}^{\text{op}}$, the limit and colimit cones exchange roles, but the idea I persists as an object in both. This is a precise sense in which ideas have symmetry: they are invariant under the reversal of arrow direction at the level of the universal properties they carry, even though the cones themselves are distinct.

13.6 Abelianization and Fidelity

Some ideas have essentially commutative expression diagrams: the order of expressions does not matter, and the cone structure is symmetric in the usual sense. Others are fundamentally non-commutative: the path through the diagram matters, and different orderings yield genuinely different objects.

When the non-commutative content is not essential to the idea—when the abelianized version is still faithful—one may work with the symmetrized structure and apply the full toolkit of homological and abelian category theory. This is a pragmatic move, analogous to passing from homotopy groups to homology: some information is lost, but what remains is tractable and meaningful.

This suggests a natural classification:

Definition 13.17. An idea I is *essentially commutative* if the abelianization of its expression diagram is faithful to I under the operations of interest. Otherwise, I is *essentially non-commutative*, and its interesting content lies in the non-abelian part.

13.7 Vacuous Cone Structure and Inclusive Design

A key feature of the framework is that cone structure exists vacuously for any idea, including bad ideas, degenerate ideas, and ideas that resist expression.

Proposition 13.18 (Vacuous Existence). *Under the Pointability Axiom, every idea I admits at least a trivial cone structure: the diagram consisting of a single pointing morphism. The limit and colimit over this one-object diagram both exist and are consistent with I .*

The richness of an idea is then measured not by whether its cone exists, but by the *density* of the cone: how many expressions populate it, how many downstream instantiations it generates, how persistent it is across media and time.

Remark 13.19 (Persistence). Ideas with rich, dense cone structures in both directions are more natural and more persistent. They can be expressed many ways, generate many downstream works, and survive across symbolic media—verbal, visual, mathematical, gestural. Ideas with thin or degenerate cones fail to be produced, fail to accumulate expressions, and do not persist. This is not a failure of the framework but a structural explanation of why some ideas endure and others do not.

13.8 Observation as Pre-Art

A further consequence of the cone structure is that observation is not passive. When a viewer encounters an artwork, they receive morphisms from the colimit cone and potentially generate new maps—new expressions, new interpretations—that enter the diagram on the limit side. In this sense:

Proposition 13.20 (Observation as Pre-Art). *The act of observation, when recorded or expressed in any symbolic form, constitutes a pre-art. Observation is a substrate transformation: it takes the substrate state of the viewer-before-encounter to the substrate state of the viewer-after-encounter, mediated by the work. This transformation is a morphism in $\mathcal{P}$.*

Painting, in particular, is simultaneously an expression and an ostension. To paint something is to point at it—to say “here, this thing”—while also transforming a substrate. Every artwork is at once an element of the colimit cone (generated from an idea) and a pointing map (a morphism into a new idea). The cone structure is not imposed on art from outside; it is what art naturally does.

13.9 Symbolic Mediation and Approximation

All expressions in the cone are symbolic: composed of primitive marks, gestures, or inscriptions with grammar, whether verbal, visual, or mathematical. There is no expression outside the symbolic. This does not restrict the framework but characterizes the mode of access:

Remark 13.21. For any idea that resists a single symbolic medium, the cone is populated by expressions across multiple media simultaneously. No single expression need be fully faithful; the idea is approximated by the intersection of all its expressions. Triangulation across symbolic systems is the natural epistemic strategy when no single map is sufficient.

Ideas that lie partially outside conscious access—subconscious or pre-linguistic ideas—are still in the diagram. They participate in morphisms. We know them through their relations to more accessible objects, not through direct introspection. The framework accommodates them without claiming to analyze them fully.

13.10 Natural Inclusivity

The preceding results establish that the Pre-Arts framework is naturally inclusive.

Theorem 13.22 (Inclusivity). *Every idea, including those that are inarticulate, subconscious, or only partially accessible, is included in the framework via the Pointability Axiom and vacuous cone structure. The framework does not require full legibility as a condition of membership.*

This is not a weakness of the theory but a design feature. A framework that could only include what it fully understood would be a very small framework. By admitting objects that participate in the diagram with thin or opaque cone structure, the theory remains large, honest about its own limits, and compatible with the full range of human creative and ideational activity.

Remark 13.23 (Ethical Analogy). The mathematical virtue and the ethical virtue coincide. A well-designed system need not fully understand every member in order to include them. Membership is granted by participation in the diagram—by the existence of morphisms—not by transparency of internal structure. This is a principle of inclusive design that the mathematics makes precise.

13.11 Summary

We have established the following:

1. Ideas have a genuine categorical geometry: cone structure, direction, universal properties, and local tangent-like structure.
2. Every idea carries two cone orientations simultaneously, corresponding to the two directions of creative process. These are distinct and neither is privileged.
3. The Pointability Axiom—that every idea admits at least one expression—must be assumed as an axiom. It is weak, reasonable, and foundational.
4. Cone structure exists vacuously for all ideas; richness is measured by cone density and persistence.
5. Observation is a pre-art. Painting is pointing. Every artwork is simultaneously an expression and an ostension.
6. The framework is naturally inclusive, accommodating ideas it cannot fully analyze through the pointing map and vacuous cone structure.

The geometry of ideas is not imposed from outside. It falls out of taking the Pre-Arts framework seriously and asking what ideas must be if artworks are morphisms. That the answer is this rich is not a coincidence. It is what the mathematics requires.

14 Pre-Arts as a Homotopy Type

14.1 Introduction

The collection of pre-arts $F(P)$ has thus far been treated as a nonabelian free group with additional categorical structure. We now ask a more refined question: does $F(P)$, or a naturally occurring subcollection of it, constitute a type in the sense of Homotopy Type Theory? The answer depends critically on the choice of equivalence relation, and there is no single canonical choice. We develop three natural notions of equivalence, each giving a distinct HoTT type with distinct geometry, and examine the type-theoretic rules for each.

A HoTT type is specified by four rules: a *formation rule* (what data is required for the type to exist), *introduction rules* (how terms of the type are constructed), *elimination rules* (how terms of the type are used), and *computation rules* (β -reduction, specifying what happens when the eliminator meets a constructor). We also examine the *uniqueness principle* (η -expansion), which asserts that every term is definitionally equal to one produced by the constructors; this does not hold for all types.

14.2 Equivalence as a Choice

Before stating the type rules, we fix the equivalence structure. A pre-art $f : A \rightarrow B$ is a morphism between substrate states. Two pre-arts $f, g : A \rightarrow B$ with the same domain and codomain may be related in several ways, giving rise to different types.

Definition 14.1 (Strict Equivalence). Pre-arts $f, g : A \rightarrow B$ are *strictly equivalent*, written $f \equiv g$, if they are identical as morphisms in $F(P)$: the same word in the free group, built from the same primitive pre-arts in the same order.

Definition 14.2 (Homotopy Equivalence of Pre-Arts). Pre-arts $f, g : A \rightarrow B$ are *homotopic*, written $f \sim g$, if there exists a 2-morphism $H : f \Rightarrow g$ —a path in the type they jointly inhabit, witnessing equivalence of process without requiring identity of construction.

Definition 14.3 (Weak / Parametric Equivalence). Given a designated forgetful functor $\phi : F(P) \rightarrow F(P)$ erasing chosen data (color, medium, scale, or other parameters), pre-arts $f, g : A \rightarrow B$ are *ϕ -equivalent*, written $f \sim_\phi g$, if $\phi(f) \equiv \phi(g)$. Strict and homotopy equivalence are the extreme cases $\phi = \text{id}$ and $\phi = \text{full localization}$ respectively.

Each choice of equivalence yields a distinct type. We develop the type rules for homotopy equivalence as the primary case, noting where strict and parametric equivalence differ.

14.3 The Type of Pre-Arts

Formation Rule

The type **PreArt** exists given:

1. A collection of substrates $\mathcal{S}$, each with a well-defined state space $S(S)$ for $S \in \mathcal{S}$.
2. A designated collection of primitive pre-arts P , where each $p \in P$ is a morphism $p : A \rightarrow B$ with $A, B \in S(S)$ for some S .
3. The free group $F(P)$ generated by P under composition with formal inverses.

The type **PreArt** is then the type whose terms are elements of $F(P)$, equipped with the equivalence structure of Definition 5.7.

Formally: given $\mathcal{S}$ and P as above,

$$\frac{\mathcal{S} \text{ type} \quad P : \mathcal{S} \rightarrow \mathcal{S}}{\text{PreArt}(\mathcal{S}, P) \text{ type}}$$

Introduction Rules

Terms of **PreArt** are constructed in three ways:

1. **Primitive introduction.** For each primitive $p \in P$ with $p : A \rightarrow B$, there is a term $\text{prim}(p) : \text{PreArt}$.
2. **Composition introduction.** Given terms $f, g : \text{PreArt}$ such that the codomain of f equals the domain of g , their composite $g \circ f : \text{PreArt}$ is a term. Composition is associative and noncommutative.
3. **Identity introduction.** For each substrate state A , the identity morphism $\text{id}_A : A \rightarrow A$ is a term $\text{id}_A : \text{PreArt}$.

Under homotopy equivalence, there is an additional constructor:

4. **Homotopy introduction.** Given terms $f, g : \text{PreArt}$ with the same domain and codomain, and a 2-morphism $H : f \Rightarrow g$, the term $\text{htpy}(f, g, H) : f =_{\text{PreArt}} g$ witnesses their equivalence. Higher homotopies are introduced analogously, giving the full ∞ -groupoid structure of Theorem 5.8.

Elimination Rules

The eliminator for **PreArt** expresses the induction principle: to define a function out of **PreArt**, it suffices to specify its values on primitives and verify compatibility with composition and homotopy.

Formally, given a type family $E : \mathbf{PreArt} \rightarrow \mathcal{U}$ (a property or structure to be assigned to each pre-art), a function $\text{rec}_{\mathbf{PreArt}} : \mathbf{PreArt} \rightarrow E$ is constructed by providing:

- For each primitive $p \in P$: a term $e_p : E(\text{prim}(p))$.
- For composition: given $e_f : E(f)$ and $e_g : E(g)$, a term $e_{g \circ f} : E(g \circ f)$ compatible with associativity.
- For homotopies: given $H : f \sim g$ and $e_f : E(f)$, a transport $H_*(e_f) : E(g)$ along H .

This is precisely what one does in practice: to assign a design principle to every pre-art in a context, it suffices to assign it to primitive pre-arts and verify that composition and homotopy preserve the principle.

Computation Rules (β -reduction)

The computation rules specify that applying the eliminator to a constructor returns the specified data:

$$\text{rec}(\text{prim}(p)) \equiv e_p$$

$$\text{rec}(g \circ f) \equiv e_{g \circ f}(e_f, e_g)$$

$$\text{rec}(\text{id}_A) \equiv e_{\text{id}_A}$$

For homotopies, the computation rule is:

$$\text{rec}(\text{htpy}(f, g, H)) \equiv H_*(\text{rec}(f))$$

That is, eliminating a homotopy witness produces the transport of the value at f along H to the value at g .

Uniqueness Principle (η -expansion)

The uniqueness principle asserts that every term of **PreArt** is definitionally equal to one built from the constructors. Whether η holds depends on the equivalence chosen.

- **Strict equivalence:** η holds. Every element of $F(P)$ is, by definition, a finite word in the generators and their inverses. The constructors (primitive introduction, composition, identity) suffice to build every term, and no term escapes this description.
- **Homotopy equivalence:** η does *not* hold in general. Two terms f and g may be homotopic—equal in the type—without being definitionally equal as words in $F(P)$. The type identifies them propositionally (via a homotopy witness) but not definitionally. This is the standard situation in HoTT: propositional equality is weaker than definitional equality, and η fails for types where the path structure is nontrivial.
- **Parametric equivalence:** η holds modulo the forgetful functor ϕ . Every term of the ϕ -equivalence type is definitionally equal to $\phi(f)$ for some constructor-built f . The uniqueness is relative to the parameter being forgotten.

The failure of η under homotopy equivalence is not a defect. It reflects the genuine content of the homotopy: two artmaking processes that produce equivalent results are identified in the type, but the witness to their equivalence—the homotopy H —carries information about *how* they are equivalent. This information is not collapsed by η and is available for use by the eliminator.

14.4 Multiple Types from One Collection

The collection $F(P)$ does not determine a unique HoTT type. It determines a family of types, indexed by the choice of equivalence:

$$\text{PreArt}_\equiv, \quad \text{PreArt}_\sim, \quad \text{PreArt}_{\sim_\phi}$$

each with its own formation, introduction, elimination, and computation rules, and each with its own geometric content.

Remark 14.4. This is not a weakness of the framework but an accurate reflection of artistic practice. Whether two pre-arts are “the same” depends on what one cares about: the same physical gesture (strict), the same process up to continuous deformation (homotopy), or the same structure ignoring medium (parametric). The theory provides vocabulary for each judgment without mandating one.

Remark 14.5 (Relation to Contexts). Every type $\text{PreArt}_\sim$ is a context in the sense of Definition 6.1: a subset of $F(P)$ with additional structure. The converse does not hold—not every context carries the coherent path structure required to be a well-formed HoTT type (Definition 6.3). Types are the well-formed contexts; the type rules above are precisely what “well-formed” means at the level of pre-arts.

14.5 The ∞ -Groupoid of Pre-Arts

Under homotopy equivalence, $\text{PreArt}_\sim$ is an ∞ -groupoid (Theorem 5.8). The type-theoretic structure developed above makes this explicit:

- **0-cells:** terms of PreArt , i.e., elements of $F(P)$.
- **1-cells:** homotopies $H : f \sim g$, i.e., paths in the type.
- **2-cells:** homotopies between homotopies, i.e., 2-morphisms $\alpha : H \Rightarrow K$.
- **n -cells:** inductively, homotopies between $(n - 1)$ -cells.

Composition of n -cells is defined inductively from the composition introduction rule. Associativity holds up to homotopy at each level, with the coherence data being the higher cells themselves. This is the standard construction of an ∞ -groupoid from a HoTT type, and it applies here without modification.

By the Homotopy Hypothesis (Grothendieck), $\text{PreArt}_\sim$ is equivalent, as an ∞ -groupoid, to a topological space: the *space of artmaking processes*. The homotopy type of this space—its fundamental group, higher homotopy groups, homology—is an invariant of the collection of pre-arts and the chosen equivalence, independent of any particular presentation.

15 Relation to Existing Work

Several prior efforts apply mathematical formalism to art theory. We note the key distinctions.

Topos-theoretic approaches treat artworks as objects with morphisms between them representing shared themes or influence. This assumes a pre-theoretic notion of artwork as object. Our framework inverts the direction: pre-arts are the morphisms, and artworks are compositions of morphisms. The inversion dissolves the foundational ambiguity about what counts as an artwork, since the theory operates below that threshold.

cARTegory Theory (Mazzola et al., 2019) applies categorical structure to the aesthetics of music and visual art. Works are again taken as objects, with the categorical structure built on top. The pre-theoretic assumption about what constitutes a work is not interrogated.

Artist as Formal System (TripleAmpersand, 2022) correctly identifies that “work comes out of work” (Serra) as a formal concatenation and gestures toward a general theory. The observation is sound but is not developed at the level of primitives, free groups, or typing judgments.

The shared failure mode is the attempt to apply formal structure to objects already called art, inheriting all the definitional instability of that term. We do not define art. We define pre-art, note that art (wherever found) is composed of pre-art, and derive the structure of that composition. The theory makes no claim about the art/non-art boundary and is therefore not falsified by any artwork, however provocative.

16 Summary of Definitions

Term	Definition
Substrate	Any material or medium subject to transformation
Conceptual substrate	An idea or semantic domain as substrate
Initial state	Least-transformed available configuration of a substrate
Pre-art	A morphism $f : A \rightarrow B$ between substrate states
Idea-arrow	A pre-art acting on a conceptual substrate
Primitive pre-art	An irreducible generator; cannot be further decomposed
Composition	Concatenation of pre-arts in $F(\mathcal{P})$; nonabelian; associative
$F(\mathcal{P})$	Nonabelian free group of all pre-arts under composition
Type	HoTT type whose terms are elements of $F(\mathcal{P})$, with path structure
Idea	Cone over all instantiations; design principles are a special case
Design principle	A cone collecting all compositions exhibiting a structural property
Context	Any subset of $F(\mathcal{P})$; constituted by its pre-arts
Well-formed context	A context carrying HoTT type structure; coherent path/equivalence structure
Universally principled context	A context admitting a map $\phi : \mathcal{C} \rightarrow \mathcal{D}$; every element exhibits a design principle
Exhibition	A context whose substrate includes a physical or virtual space
Local coherence	All compositions in a sub-context share a common type
Universal pre-art	Initial object of a context; maps uniquely to every other element via set-theoretic functions
Universal expression-ism	The regime of universally principled contexts
Homotopy	2-morphism $H : f \Rightarrow g$; path witnessing equivalence of processes